Question I: Some information about four metals

This more or less long task deals with four metals which play a very important role in technology. They are not too far away from each other in the periodic table of elements. For each of these metals there is a set of problems which is connected with analytical and/or physical chemistry.

Which metals are we talking about?

In order to answer this question there is a series of information available:

- The sum of the molar masses of the metals equals about 227 g.
- A very important oxide of metal ① contains ≈ 40 % oxygen by mass, and is the most important white pigment in industry.
- Metal ① is also used in airplane construction.
- Metal ② occurs in many compounds with different oxidation numbers. The compounds and their aqueous solutions respectively show the following colours:
- Me(+2).....colourless
- Me(+3).....light brown
- Me(+4).....black
- Me(+6).....green
- Me(+7).....violet
- Metal ③ exhibits ferromagnetism at room temperature.
- Some people react allergic when coming into skin-contact with metal ③.
- Metal ④ is the only which produces a colourless, very poorly soluble sulphide. ($K_{SP} = 2 \cdot 10^{-24}$). 1.375 ng of the sulphide dissolve in 10 L of water.

1.1. Write the formulae and the names of the metals ① to ④.

1.2. Give reasons for your assumptions by calculations wherever this is possible.

Concerning metal ①

Metal ① is today produced on big scale using the method by W. Kroll (1932). Thereby the tetrachloride of the metal is reduced with magnesium. The tetrachloride is gained from one of the most important ores of this metal Fe①O_3 by reaction with chlorine and carbon at 900°C. The metallic sponge produced by this method is purified using aqua regia (3 mol HCl + 1 mol HNO_3) and then molten to bars under argon. The annual world production amounts to more than 100000 t.

1.3. Write down a balanced equation for the production of the tetrachloride. Aside from the tetrachloride iron(III) chloride and carbon oxide are produced.

1.4. Write down a balanced equation for the Kroll-process.

1.5. In aqua regia, chlorine, NOCl and water are produced from HCl and HNO_3. Write down a balanced equation for this reaction.

1.6. If we start with 1.0 t of the ore in question, which mass of metal ① can be produced. The yield in the first process is 85%, and in the Kroll-process 95 %. Show your calculation.

1.7. What is the volume of the necessary chlorine amount in part 1.3. for 1.0 t of ore, if a stoichiometric proportion is chosen? (p = 1.0 bar)? Show your calculation.

Concerning metal ②

Metal ② is a "star" of redox chemistry.

It is possible to oxidize a lot of materials using one of the metal species in acidic solution. This species ($②O_4^-$) is thereby converted to $②^{2+}$.

It is also possible to use this species in basic solution as an oxidant, the final product however is $②O_2$.

The species $②O_4^{2-}$ disproportionates to $②O_4^-$ and $②O_2$.

1.8. Write down a balanced equation for the disproportion reaction.

1.9. Write down a balanced equation for the reaction of $②O_4^-$ with $C_2O_4^{2-}$ in acidic surrounding.

1.10. Write down a balanced equation for the reaction of $②O_4^-$ with BaO_2 in basic surrounding.

1.11. 15.3 mL of a solution of $②O_4^-$ (c = 0.0200 M) were used to titrate potassium oxalate monohydrate in acidic solution. Calculate the mass of the potassium salt in the sample.

Concerning metal ③

Metal ③ crystallizes forming a cubic face centred lattice, and has the density 8.91 $g \cdot cm^{-3}$.

1.12. How many atoms of ③ are found in a unit cell of the element?

1.13. Calculate the length of the lattice constant.

1.14. The maximum of the 2nd-order reflection of the [111]-plane in the unit cell of ③ will occur at an angle of 20.38°. Calculate the energy of the X-radiation which as used for the recording, using the unit keV.

Concerning metal ④

The positively charged (2+) ion of metal ④ forms a hexaquo complex in aqueous solution. One of the H_2O-ligands is exchanged against OH^- in an extremely rapid reaction.

The kinetical data of this reaction (c_0(aquo complex) = $c_0(OH^-)$ = 0.0050 mol/L) were investigated through laser supported kinetical analysis in the nanosecond region.

t(ns)	40	80	120	160	200
c_t (mmol/L)	3.57	2.78	2.27	1.92	1.67

1.15. Write down a balanced equation for the ligand exchange reaction.

1.16. Calculate the reaction order and the rate constant of the reaction.

1.17. Calculate the half life of the reaction.

As a closer: About all the metals, their oxides and their behaviour towards C and CO

C and CO act as reducers with many metal oxides, although the reductive effect is strongly temperature dependant. Additionally the reductive species are connected via the following equilibrium:

$$2\ CO \rightleftarrows CO_2 + C \qquad \Delta_R H^0 = -172.5\ kJ \text{ and } \Delta_R S^0 = -176.5\ J/K$$

1.18. At which temperature is the equilibrium shifted from the left to the right side (or vice versa), if you assume that the caloric data are independent of the temperature? Show by a calculation.

It is possible to show that among all the different pressure conditions there is a temperature interval where CO_2 decomposes directly into C and O_2, without formation of CO. This happens for 1 bar and 400°C:

$$CO_2 \rightleftarrows O_2 + C \quad \Delta_R H^0 = 283.3\ kJ \text{ and } \Delta_R S^0 = -13\ J/K$$

1.19. Where is the equilibrium at the given conditions? Show by a calculation.

1.20. At which values of the reaction quotient will the system react towards the right side? Show by a calculation.

Question II: Sulfur determination

Compounds of sulfur in its lower oxidation states are present in many industrial wastes (metallurgy, production of paper, chemical) and are dangerous ecotoxicants. The prevalent forms of sulfur in lower oxidation states in solutions are S^{2-}, SO_3^{2-} and $S_2O_3^{2-}$ ions. Their content can be determined by redox titration under different conditions.

2.1. To a 20.00 mL sample containing S^{2-}, SO_3^{2-} and $S_2O_3^{2-}$, an excess of $ZnCO_3$ suspended in water was added. Upon completion of the reaction the solution was filtered into a 50.00 mL volumetric flask and diluted to the mark. To 20.00 mL of the filtrate an excess of aqueous formaldehyde was added. The mixture was acidified with acetic acid and titrated with 5.20 mL of 0.01000 M standard solution of iodine.

a) Write down the net ionic equations of the reactions taking place during the analysis.

b) Which ion, S^{2-}, SO_3^{2-} and $S_2O_3^{2-}$, can be determined by this method?

c) Calculate the concentration of this ion in ppm in the initial solution.

2.2. A 20.00 mL sample of the 0.01000 M iodine solution was acidified with acetic acid and then combined with 15.00 mL of the filtrate above. The mixture was titrated with 6.43 mL of the 0.01000 M sodium thiosulfate standard solution.

a) Write down the net ionic equations of the reactions taking place during the analysis.

b) Which ion, S^{2-}, SO_3^{2-} and $S_2O_3^{2-}$, can be determined by this method taking into account the result of the previous experiment?

c) Calculate the concentration of this ion in ppm in the initial solution.

2.3. A 10.00 mL sample of 0.05000 M iodine solution was acidified with acetic acid and then 10.00 mL of the original sample containing S^{2-}, SO_3^{2-} and $S_2O_3^{2-}$ were added. The mixture was titrated with 4.12 mL of 0.05000 M sodium thiosulfate standard solution.

a) Write down the net ionic equations of the reactions taking place during the analysis.

b) Which ion, S^{2-}, SO_3^{2-} and $S_2O_3^{2-}$, can be determined by this method taking into account the results of two previous determinations?

c) Calculate the concentration of this ion in ppm in the initial solution.

Question III: Spinel Oxides

The simple d-block oxides such as Fe_3O_4 and Co_3O_4 and many related mixed metal compounds have important properties. They have structures related to the mineral spinel, $MgAl_2O_4$, and may be given a general formula of AB_2O_4.

Stoichiometric amounts of two aqua complexes of transition metal (**A** and **B**) nitrate salts are thermally reacted to form a spinel AB_2O_4 crystalline solid that has a face-centered cubic (*fcc)* structure with a unit cell composition of 8 AB_2O_4. Depending on the location of these two cations (A and B), the spinel structures are divided into two categories as normal and inverse spinels. In a normal spinel, the A^{2+} ions occupy the tetrahedral holes and the B^{3+} ions occupy the octahedral holes, but in the inverse spinel structures, the 2+ ions are replaced by half of the 3+ ions in the structure.

Crystalline solid has an ordered structure in which the unit cell repeats along all 3 principal axes of a three-dimensional matter. The smallest group of atoms in the material that constitute this repeating pattern is the unit cell of the structure. The unit cell completely reflects the geometry and structure of the entire crystal, which is built up by repetitive translation of the unit cell along the principal axes. Face centered cubic (*fcc*) is one of a common structure type of crystalline solid. Anions (X) are in the corners and faces of a cube (1/8 from each corners and ½ from each faces, because the corners and faces are shared by 8 and 2 unit cells, respectively) in the simplest *fcc* structure. The cations (M) occupy the holes among the anions. There are 8 tetrahedral (corners) and 4 octahedral holes (1 at the middle and 3 on the edges, each edge has

¼ octahedral hole) in a *fcc* structure. Therefore, the unit cell composition is M_4X_4 with an empirical formula of MX. However, the unit cell of a spinel structure is constructed by 8 of these *fcc* units.



29.746 g salt of **A** was mixed with 58.202 g salt of **B** in a thermal process to produce 24.724 g pure product, AB_2O_4. In the spinel formation process, the metal ion of salt A keeps its oxidation state but the metal ion **B** undergoes oxidation. Both salts contain the same number of the water molecule(s), metal ion, and nitrate ion(s). Elemental analysis of the spinel provided the following data: 6.538 g metal A and 11.786 g metal B. Assume the end product is a diamagnetic solid matter. Considering the information provided above. Answer the following questions.

3.1. Suggest formulas for the salts of **A** and **B**.

3.2. Draw the structure of one of the complex ions i) without and ii) with one of the nitrates being in the coordination sphere as a bidentate ligand and **identify** if the inversion center is present in the complexes. Inversion is a symmetry operation that translates every atom through the center to the opposite side.

3.3. Place the metal ions in an appropriate location in the crystal structure and **suggest** if it is a normal or inverse spinel.

The x-ray diffraction data of AB_2O_4 provides a unit cell parameter of 8.085 Å, which is constructed from 8 *fcc* units and corresponds to a length of the edges of the cube.

3.4. Sketch one of the *fcc* units of AB_2O_4 and **place** the atoms in the unit.

3.5 What is the density of AB_2O_4? (hint: 1 Å is 1.0×10^{-10} m)

Reacting this spinel with other transition metals (M) produces M doped AB_2O_4, where M has a choice of occupying the place of either A or B-sides. The side product is **AO (mono-oxide of A)**.

3.6. M is Mn^{2+} in compound **C** and Ni^{2+} in compound **D**, **suggest** the location of Mn^{2+} and Ni^{2+} ions in the structure of **C** and **D**, respectively. Assume splitting energy in Ni^{2+} and B^{3+} are 11500 cm^{-1} and 20800 cm^{-1} in the octahedral field, respectively, and the pairing energy is 19500 cm^{-1}.

If the doping is in a small quantity or in some cases the doped metal ion behaves like a free ion in the lattice (it means, electrons of M only feel the surrounding atoms and localized to M and its 1 shell of atoms in the structure). Assume Mn^{2+} is behaving like a free ion in the lattice and creating its own localized electronic energy levels.

3.7. <u>Draw</u> the d-orbital splitting and identify if the Mn^{2+} species are paramagnetic or diamagnetic.

Magnetic susceptibility could be calculated from the spin only formula:

$$\mu(spin\ only) = (n(n+2)^{1/2},$$

where n is a number of unpaired electrons. However, some other electronic couplings affect the magnetic moment such that a correction term is needed. The correction term α is related to ground state (where α = 4 for a non-degenerate and 2 for a degenerate ground state, degeneracy of a ground state can be determined from the electron configurations, such as completely filled and half-filled set of orbitals creates a non-degenerate and a partially filled set of orbitals create degenerate states) and $\lambda = 88\ cm^{-1}$ for Mn^{2+} and -315 cm^{-1} for Ni^{2+}), and splitting energy (Δ is 5000 cm^{-1} for Mn^{2+} and 11500 cm^{-1} for Ni^{2+}) and the magnetic moment is:

$$\mu_{eff} = \mu(spin\ only)\left(1 - \frac{a\lambda}{\Delta}\right).$$

The magnetic susceptibility can be experimentally determined and it is interrelated with the magnetic moment (if we ignore the diamagnetic contributions) with the following formula:

$$\mu_{eff} = 2.828(X_m T)^{1/2},$$

where T is temperature in Kelvin and X_m is the molar magnetic susceptibility.

3.8. <u>What is</u> the magnetic susceptibility of the products at 25 °C, if the samples C and D weigh 25.433 and 25.471 g, respectively (each obtained from 24.724 g AB_2O_4)?

3.9. <u>Place</u> all the metal ions (A, B, Mn^{2+}, and Ni^{2+}) into their appropriate locations in the lattice and **<u>fill up</u>** the following table. Use t_{2g} for d_{xy}, d_{xz}, and d_{yz} and e_g for d_{x2-y2}, d_{z2} orbitals in octahedral (O_h) and t_2 and e orbitals in tetrahedral (T_d) cases. If there is distortion, **<u>predict</u>** the type of distortion(s) and **<u>show</u>** the d-orbital splitting.

M	**Local geometry**	**Electron configuration**	**Degeneracy**	**Type of distortion**